

SHAYAN BHOWMIK

CSE Undergraduate

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Summary

Second-year Computer Science student at Symbiosis Institute of Technology with hands-on experience in software development, system design, and problem-solving. Actively builds practical skills through personal projects and hackathons, applying core programming concepts to real-world challenges. Eager to explore diverse domains within computer science, collaborate with like-minded peers, and contribute to innovative, tech-driven environments.

Skills

Programming & Querying: C++, Java, Python, Javascript, MySQL

Other Skills: Blockchain, Quantum Computation

Soft Skills: Leadership, Problem Solving, Adaptability, Decision Making

Experience

GirlScript Summer of Code (GSSoC)

April 2026 – Present

Open Source Contributor (Remote)

- Contributing to open-source projects as part of a structured program, collaborating with developers across the globe on shared codebases.
- Submitted pull requests addressing bugs, feature additions, and improvements, following standard open-source contribution workflows including code review and version control.
- Reviewed peer contributions and improved project documentation to support better usability and contributor onboarding.
- Communicated and coordinated with maintainers and fellow contributors through async channels, gaining practical experience in collaborative software development.

Projects

Image Processing Brain MRI | *Python*

January 2026 – April 2026

- **Developed an AI-assisted Brain MRI analysis system** to detect abnormal neurological patterns using 3D NIfTI MRI data.
- Designed a preprocessing pipeline to convert volumetric MRI scans into model-ready slices with normalization and modality-specific handling (FLAIR).
- Implemented segmentation-based labeling for tumor detection (BRATS) and integrated healthy control data (OASIS) to enable true abnormal vs normal classification.
- Integrated explainable AI using Grad-CAM to highlight critical brain regions influencing model predictions, improving clinical interpretability.

Post-Quantum Risk Modeling Framework | *Python, Pandas, Numpy*

March 2026 – Present

- **Built a quantum-crypto impact simulation framework** to analyze how quantum computing affects the security of modern cryptographic systems like RSA, SHA-256, and ECC.
- Modeled classical and quantum attack complexities by implementing Grover's and Shor's algorithms, quantifying security reduction in $\log_2$ space for accurate comparison.
- Incorporated realistic quantum hardware constraints (NISQ, near-term, fault-tolerant) to evaluate practical feasibility using qubit requirements, error rates, and execution time estimates.
- Developed simulation modules to estimate time-to-break and identify critical security thresholds, revealing phase-transition behavior where RSA security collapses beyond ~ 4099 logical qubits.

Awards & Certifications

- **Hashgraph Developer – The Hashgraph Association (2025):** Blockchain bootcamp conducted by the Hashgraph and Hedera Association.
- **Introduction to Prompt Engineering – Microsoft (2025):** Understanding of prompt engineering from basic to advance level.
- **Python – HackerRank (2025):** Certified Python knowledge by HackerRank.